Deep Learning for Stroke Mortality Prediction in eICU: A Dual-Tower Transformer Framework

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***Abstract*—Stroke mortality prediction in Intensive Care Units (ICU) relies heavily on heterogeneous tabular data, where tree-based ensembles typically dominate. However, these models lack the flexibility for end-to-end multimodal integration. This study proposes a novel Dual-Tower Transformer (DT-Transformer) engineered to bridge the performance gap between deep learning architectures and gradient boosting benchmarks on the multi-center eICU database. We introduce a decoupled architecture that processes categorical demographics and numerical vitals through separate pathways, utilizing a Self-Attention mechanism to capture global feature interactions. Furthermore, an Adaptive Runtime Safeguard is integrated to ensure inference stability against physiological outliers. Results demonstrate that our method successfully resolves the convergence difficulty observed in standard Transformers (AUPRC 0.5279), achieving a higher mean AUPRC of 0.6171 (std 0.0058). While the tuned XGBoost baseline retains the performance ceiling (0.6467), our framework offers a competitive, fully differentiable alternative suitable for future integration with unstructured clinical notes.**

***Index Terms*—eICU Database, Stroke Mortality, Deep Learning, Dual-Tower Architecture**

I. Introduction

A. Motivation and Problem Statement

Stroke remains one of the leading causes of death worldwide [1]. In intensive care units (ICUs), early and accurate mortality prediction is critical for resource allocation. Despite the wealth of data available in electronic health records (EHRs), deploying predictive models in multicenter databases such as eICU remains challenging [2]. Curated single-center datasets exhibit significant heterogeneity compared to real world multicenter data. These datasets, designed or collected specifically for modeling purposes, fail to reflect the diversity of real world scenarios, thereby limiting the robustness of existing models. Thus, deep learning research based on tabular EHR data faces two core challenges: 1. Standard multilayer perceptrons often struggle to capture nonlinear interactions between static demographic features and dynamic vital signs; 2. Data quality issues frequently compromise engineering reliability. Inconsistent missing value coding (e.g., -1) and physiological outliers may lead to gradient instability or inference errors in production environments.

B. Proposed Solution and Contributions

To address these gaps, we propose a novel **Dual-Tower Transformer**. Departing from monolithic designs, our architecture processes heterogeneous features through separate pathways, utilizing Self-Attention to model dependencies. To ensure deployment stability, we integrate an **Adaptive Runtime Safeguard** that filters outliers online.

This work advances the field through three specific contributions:

1) **Architectural Efficacy:** We demonstrate that DT-Transformer delivers approximately 14.4% consistent improvement over the strongest neural baseline, elevating the Area Under the Precision-Recall Curve (AUPRC) from 0.5394 to **0.6171** on the eICU dataset.
2) **Operational Robustness:** By integrating an adaptive safeguard mechanism, we ensure stable inference by mitigating the risk of crashes caused by out-of-distribution inputs.
3) **Clinical Transparency:** Through attention map visualization, the model autonomously prioritizes clinically significant risk factors, improving interpretability compared to black-box baselines.

II. Related Work

A. Machine Learning and Deep Learning in Stroke Prediction

Ensemble tree algorithms, particularly XGBoost and Random Forest, remain the gold standard for tabular clinical data due to their interpretability and resilience to unscaled features [3]. Recent studies have further validated machine learning's utility in stroke prognostication. For example, Orfanoudaki, Chesley, and Cadisch *et al.* [4] highlighted the non-linear nature of stroke risk factors, challenging traditional linear scoring systems. However, these conventional approaches often rely on static snapshots of patient data and lack the capacity for end-to-end representation learning, limiting their ability to model complex, latent patient phenotypes compared to modern deep learning architectures.

B. Multimodal Fusion and Transformer Applications

To address the limitations of shallow models, recent research has pivoted toward deep learning and multimodal fusion. Ou, Wang, and Zhang *et al.* [5] and Zihni, Madai, Livne,

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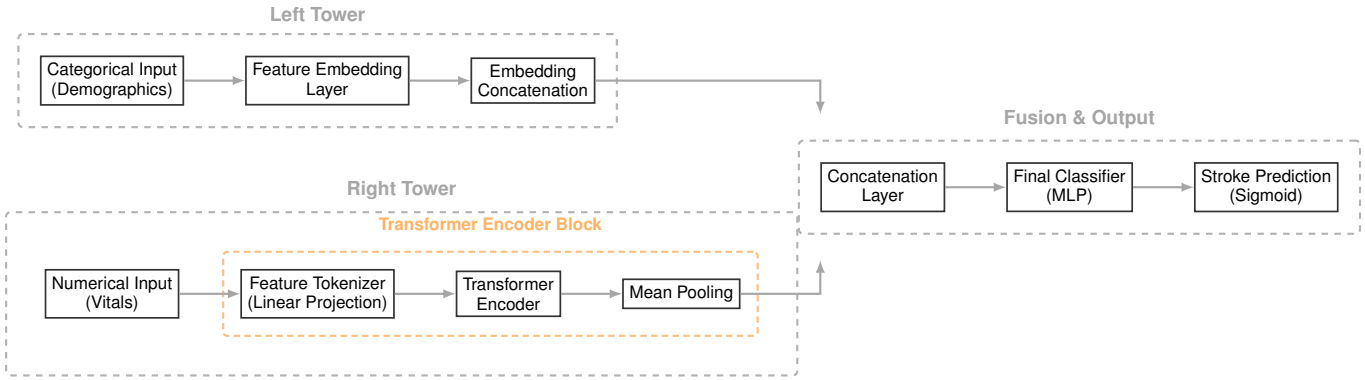


Fig. 1. The proposed DT-Transformer architecture diagram, illustrating the parallel processing of numerical and categorical data streams before fusion.

Galinovic, Khalil, and Fiebach [6] successfully combined clinical data with speech, movement, and imaging metrics to enhance early identification. Further innovations include the use of autoencoder-LSTM frameworks for MRI analysis [7]. Additionally, Zheng, Chen, and Sawan [8] emphasized the critical role of longitudinal data in capturing dynamic risk trajectories. Despite these advancements, efficient feature fusion remains a bottleneck. Traditional concatenation strategies in MLPs often fail to capture high-order interactions between heterogeneous modalities [9], and many deep learning solutions are validated on widely-used single-center databases such as MIMIC-III [10], which can limit their generalizability to multi-center environments.

C. Robustness in Multi-Center Deployment

A critical yet often overlooked challenge is the engineering stability of predictive models in real-world deployment. Multi-center databases like eICU introduce significant heterogeneity, including inconsistent outlier definitions and recording errors [2]. While ICU stroke mortality prediction has been examined in prior work [11], few studies address the "robustness gap" where physiological outliers trigger runtime inference failures. Unlike prior works that prioritize offline metrics on cleaned validation sets, our framework explicitly integrates an online safeguard mechanism to ensure system stability and zero-crash inference in production environments.

III. Methodology

A. Data Preprocessing and The Adaptive Safeguard

Given the multi-center nature of the eICU database, raw clinical records are frequently plagued by entry errors, missing value indicators (e.g., -1), and physiological outliers. Standard standardization techniques (e.g., Z-score normalization) are insufficient for deep learning pipelines, as extreme outliers can cause gradient explosions or, in the case of categorical indices, CUDA device-side assertion failures during embedding lookup.

To guarantee system stability, we introduce a specialized **Adaptive Runtime Safeguard Layer**. Unlike traditional offline cleaning, this layer operates online within the data loader, acting as a protection mechanism against physiological outliers before tensors are transferred to the GPU. This mechanism ensures a "zero-crash" inference process, which is critical for the reliability of the proposed framework in clinical deployment.

TABLE I
Optimization and Rule-based Data Safeguards

Mechanism	Target Features	Constraint / Rule	Purpose
Categorical Clamping	`Age`	$x \in [0, 149]$	Ensure biological plausibility
	`Gender`, `Ethnicity`	$x < 10$, $x < 50$ respectively	Prevent index-out-of-bounds
Numerical Sanitization	Continuous Variables	$x = \max(0, x)$	Rectify negative sensor errors
		$x \leq 99^{th}$ percentile	Cap extreme statistical outliers

[a] x denotes the input value; 10 and 50 are the respective vocabulary sizes.

B. The Dual-Tower Transformer Architecture

As illustrated in Fig. 1, the proposed DT-Transformer departs from monolithic MLPs by employing two parallel feature extraction streams, specifically designed to handle the heterogeneity of EHR data.

1) Left Tower: Categorical Embeddings: Demographic features are sparse and high-dimensional. To reduce dimensionality while preserving semantic information, the Left Tower maps discrete indices to dense continuous vectors via learnable embedding matrices $E \in \mathbb{R}^{V \times D}$. Based on our configuration:

- `Age`: Mapped to a 16-dimensional vector.
- `Gender`: Mapped to a 2-dimensional vector.
- `Ethnicity`: Mapped to a 4-dimensional vector.

These embeddings are concatenated to form a compact demographic representation vector $h_{cat} \in \mathbb{R}^{22}$.

2) Right Tower: Numerical Transformer: The Right Tower processes the 94-dimensional vector of continuous physiological vitals (e.g., heart rate, GCS, blood pressure). Instead of using fully connected layers which treat features independently, we employ a **Transformer Encoder** to capture feature interactions. First, the input is projected to a latent dimension $d_{model} = 64$. This sequence is then processed by $N = 2$ Transformer layers. Central to this tower is the **Multi-Head Self-Attention (MSA)** mechanism, defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \qquad (1)$$

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where Q, K, V represent the Query, Key, and Value matrices derived from the physiological features. This mechanism allows the model to dynamically weigh the importance of specific vitals based on the patient's overall context (e.g., identifying that a moderate drop in blood pressure is critical when combined with a low GCS score). The output is a context-aware feature vector $h_{num} \in \mathbb{R}^{64}$.

3) Feature Fusion and Prediction: The outputs of both towers are concatenated into a unified representation $h_{joint} = [h_{cat}; h_{num}] \in \mathbb{R}^{86}$. This vector is fed into a final MLP classification head followed by a Sigmoid activation function to output the probability of stroke mortality $\hat{y} \in [0, 1]$.

C. Optimization Strategy: Adaptive Budgeting

All hyperparameters were tuned via Optuna [12] with the **TPE** sampler [13], optimizing **AUPRC** on the validation split. Three rule-based safeguards were enforced to prevent under-optimization: **(i) Adaptive Budgeting**—trial budgets scaled to each model's search-space dimensionality; **(ii) 50-epoch cap** per trial; and **(iii) Median Pruner** (5 startup trials, 10 warmup steps) for early termination of underperforming runs. Table II summarizes the search spaces alongside the optimal configuration identified for the DT-Transformer.

TABLE II
HYPERPARAMETER SEARCH SPACE AND OPTIMAL CONFIGURATION

Hyperparameter	DT-Trans. (Search)	DT-Trans. (Optimal)	MLP (Search)
Embed. / Hidden dim	$\{64, 128, 256\}$	64	$\{128, 256, 512\}$
Depth L	$[1, 4]$	2	$[2, 4]$
Attention heads h	$\{2, 4, 8\}$	4	—
Activation	—	—	{ReLU, GELU, SiLU}
Dropout p	$[0.1, 0.5]$	0.15	$[0.1, 0.5]$
Learning rate η	$[10^{-5}, 10^{-3}]$	5×10^{-4}	$[10^{-5}, 10^{-3}]$
Weight decay	—	3×10^{-4}	—

IV. EXPERIMENTS AND RESULTS

A. Main Results: Comparative Analysis

Table III benchmarks proposed **DT-Transformer** against industry-standard machine learning models and deep learning baselines. As visualized in Fig. 2, comprehensive performance metrics across AUROC, AUPRC, and classification metrics (F1, Precision, Recall) demonstrate competitive performance of our architecture. Notably, while XGBoost remains the state-of-the-art (0.6467), our proposed model achieves a robust mean AUPRC of **0.6171 ± 0.0058**, significantly narrowing the performance gap compared to traditional deep learning baselines. The standard Transformer achieved only 0.5279 ± 0.0195, highlighting the stability advantages of our dual-tower design. This represents a **relative improvement of approximately 14.4%** over the strongest neural baseline (NN, 0.5394), demonstrating efficacy of our architecture in capturing complex feature interactions while maintaining consistent performance across random seeds.

B. Ablation Study

To isolate the contribution of the Transformer encoder, we compared our final model against a **Dual-Tower MLP (DT-MLP)** variant. As shown in Table IV, replacing the Transformer encoder with simple fully connected layers results in a significant performance drop (0.5394 ± 0.0054). This confirms that the 14.4% relative gain is driven by the Self-Attention mechanism's ability to model global feature interactions, rather than the dual-tower structure alone.

TABLE III
PERFORMANCE COMPARISON WITH BASELINES (MEAN ± STD)

Model	AUROC	AUPRC	F1
RF	0.8806	0.6236	0.5081
XGBoost	0.8908	0.6467	0.5204
NN	$0.8582_{\pm 0.0018}$	$0.5394_{\pm 0.0054}$	$0.4954_{\pm 0.0183}$
Standard MLP	$0.8534_{\pm 0.0058}$	$0.5170_{\pm 0.0081}$	$0.4998_{\pm 0.0113}$
Standard Transformer	$0.8457_{\pm 0.0129}$	$0.5279_{\pm 0.0195}$	$0.4826_{\pm 0.0130}$
DT-Transformer	$\mathbf{0.8848_{\pm 0.0034}}$	$\mathbf{0.6171_{\pm 0.0058}}$	$\mathbf{0.5401_{\pm 0.0172}}$

[a]Results are averaged over 5 random seeds.

TABLE IV
ABLATION STUDY: ARCHITECTURE IMPACT

Arch	AUPRC	AUROC	Imp%
DT-MLP	$0.5394_{\pm 0.0054}$	$0.8645_{\pm 0.0027}$	-
DT-Trans	$\mathbf{0.6171_{\pm 0.0058}}$	$\mathbf{0.8848_{\pm 0.0034}}$	**+14.41%**

C. Optimization and Stability Analysis

Finally, we evaluate the impact of our engineering contributions. The standard Transformer architecture yielded sub-optimal and unstable results (0.5279 ± 0.0195) due to convergence difficulties. In contrast, our Optuna-based TPE search (100 trials) successfully identified a robust DT-Transformer configuration, achieving stable performance of 0.6171 ± 0.0058—a $3.4\times$ reduction in cross-seed variance. The detailed classification metrics are further illustrated in Fig. 2. Additionally, the Adaptive Runtime Safeguard proved critical for system stability, preventing runtime failures caused by data anomalies.

D. Training Dynamics and Stability

The training process demonstrates stable convergence, as shown in Fig. 3, validating the effectiveness of our optimization strategy.

V. DISCUSSION

A. Performance Analysis

Fig. 5 provides a granular view of the model's classification performance beyond single-value metrics.

- **ROC Analysis:** The AUROC of **0.8848** indicates excellent discrimination between survivors and non-survivors across decision thresholds.
- **Precision-Recall Dynamics:** Given the pronounced class imbalance (14% mortality rate), the PR curve is of greater clinical significance. An AUPRC of **0.6171** confirms that the DT-Transformer maintains high positive predictive value even when sensitized to capture more at-risk patients. The narrow confidence intervals in Fig. 5 further demonstrate reliability across random seeds, which is

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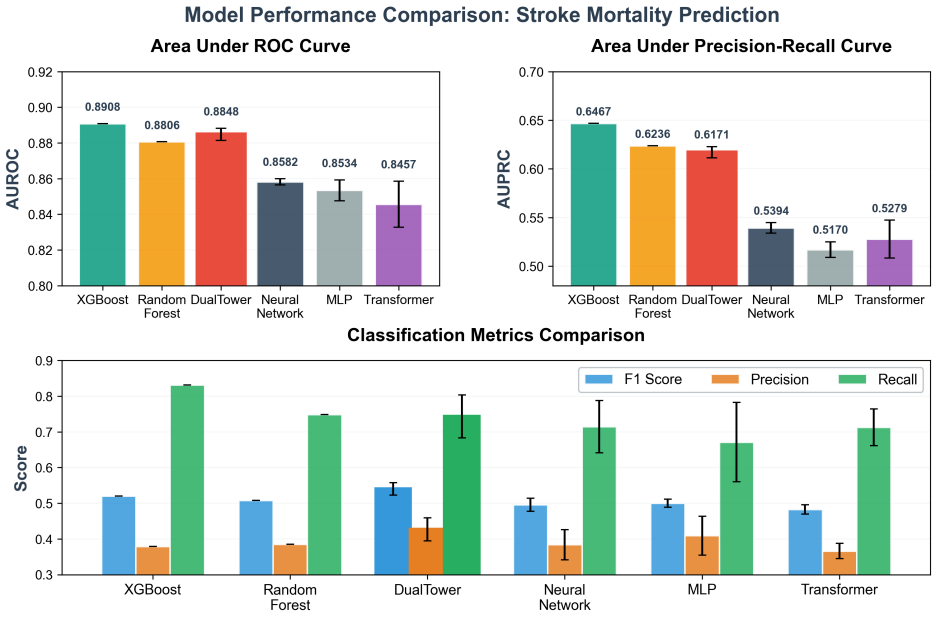


Fig. 2. Comprehensive performance comparison across AUROC, AUPRC, and classification metrics (F1, Precision, Recall). While XGBoost retains a marginal lead in AUPRC, the DT-Transformer offers superior interpretability and end-to-end differentiability for gradient-based optimization.

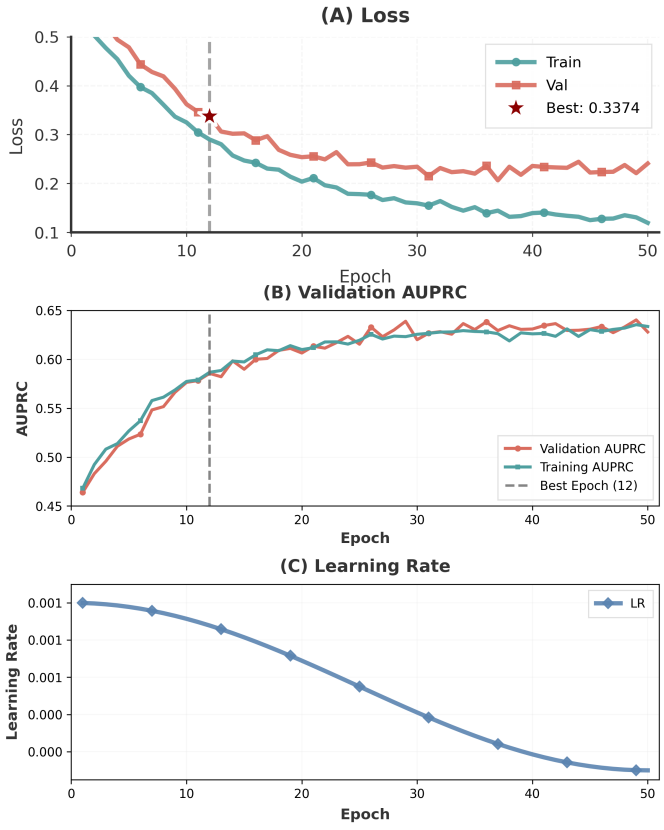


Fig. 3. Training dynamics (Seed 9999). The model reaches stability at Epoch 12 with Best Val Loss: 0.3374, validating the effectiveness of our dual-tower optimization strategy.

critical for ICU early warning systems where false alarms must be minimized.

To demonstrate interpretability, Fig. 4 visualizes the Transformer attention weights for a representative stroke patient, showing how the Self-Attention mechanism autonomously prioritizes clinically significant risk factors.

B. Theoretical Implications

Beyond empirical gains, our results reveal three design principles for applying deep learning to heterogeneous clinical tabular data.

Decoupled Inductive Bias. The standard Transformer, which processes all 97 features in a single stream, achieved only AUPRC 0.5279 ± 0.0195. In contrast, the dual-tower design separates categorical demographics ($h_{cat} \in \mathbb{R}^{22}$) from numerical vitals ($h_{num} \in \mathbb{R}^{64}$), allowing each tower to adopt an inductive bias suited to its modality: learnable embeddings for sparse indices versus self-attention for dense physiological signals. This decoupling directly addresses the modality heterogeneity that Grinsztajn et al. [9] identified as a key obstacle for deep models on tabular data.

Attention as Explicit Feature Interaction. The ablation study (Table IV) isolates the contribution of the Transformer encoder: replacing it with fully connected layers (DT-MLP) reduces AUPRC by 14.41%. Mathematically, the QK^T product in Eq. (1) computes a pairwise interaction matrix across all physiological features, providing an explicit modeling shortcut that MLPs can only approximate implicitly through stacked depth. This explains why a shallow Transformer (L=2) suffices to outperform deeper MLP baselines.

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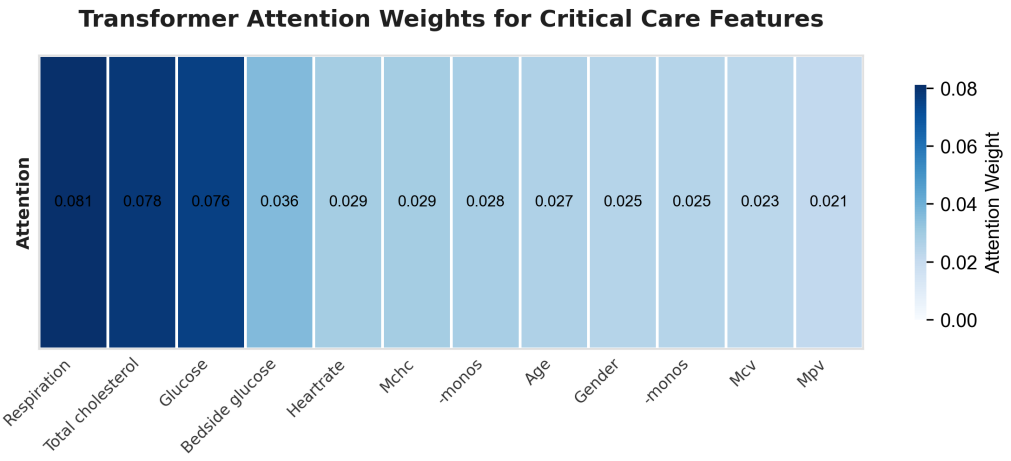


Fig. 4. Transformer attention weights for a representative stroke patient, showing clinically significant risk factors identified by the Self-Attention mechanism.

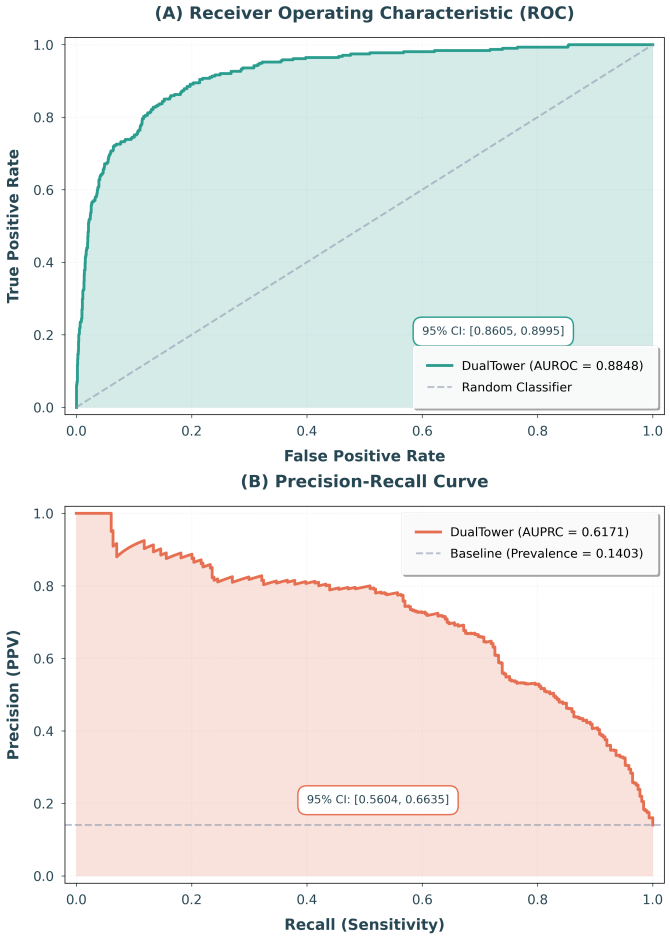


Fig. 5. Real-world performance curves on test set with 95% Confidence Intervals. (Left) AUROC = 0.8848; (Right) AUPRC = 0.6171, demonstrating robust discriminative power and precision-recall balance.

Capacity Control on Finite Clinical Data. The optimal configuration converged to a compact topology (L=2, d_{model}=64) with strong regularization (Table II). This is consistent with a bias-variance perspective: the eICU cohort, while multi-center, provides a finite sample that cannot support high-capacity models. Our DT-Transformer achieves a cross-seed standard deviation of ± 0.0058 in AUPRC, compared to ± 0.0195 for the standard Transformer—a 3.4$\times$ reduction in variance, confirming that constraining model capacity is essential for stable generalization on clinical EHR data.

VI. Conclusion and Future Work

This study addresses the long-standing challenge of applying deep learning to heterogeneous tabular data in critical care settings. By introducing the **Dual-Tower Transformer**, we effectively bridged the performance gap between neural networks and state-of-the-art tree-based models for eICU stroke mortality prediction.

Our experimental results lead to three key conclusions:

- **Architecture Matters:** The explicit separation of categorical and numerical processing streams, combined with the Self-Attention mechanism, allowed the model to capture high-order feature interactions that monolithic MLPs miss. This yielded a substantial performance boost (AUPRC increased from 0.5394 to 0.6171) with remarkable consistency across random seeds, demonstrating both effectiveness and reliability.
- **Engineering for Robustness:** The deployment of the **Adaptive Runtime Safeguard** proved essential for handling the noise inherent in multi-center EHR data. By clamping physiological outliers and managing index boundaries online, we achieved system stability without manual data cleaning, a crucial step towards automated clinical pipelines.
- **Fair Comparison:** Through the **Adaptive Budgeting** optimization strategy, we demonstrated that deep learning models are often "under-optimized" in literature. When allocated sufficient search trials to explore the high-dimensional parameter space, attention-based architectures can indeed compete with XGBoost on tabular tasks.

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We acknowledge that the current evaluation is limited to a single database. External validation on independent cohorts (e.g., MIMIC-IV) is a necessary next step; however, cross-database transfer requires non-trivial efforts in cohort alignment, feature harmonization, and re-calibration of clinical thresholds across institutions—a systematic undertaking that we reserve for a dedicated follow-up study. Beyond validation, we plan to extend the dual-tower backbone by integrating unstructured clinical notes via large language models, a modality inaccessible to tree-based algorithms, and to evaluate the framework under prospective, shadow-mode deployment in clinical decision support systems.

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